

Finding patterns in biological sequences

BiolInf 2025/2026 1st Semester

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Summary

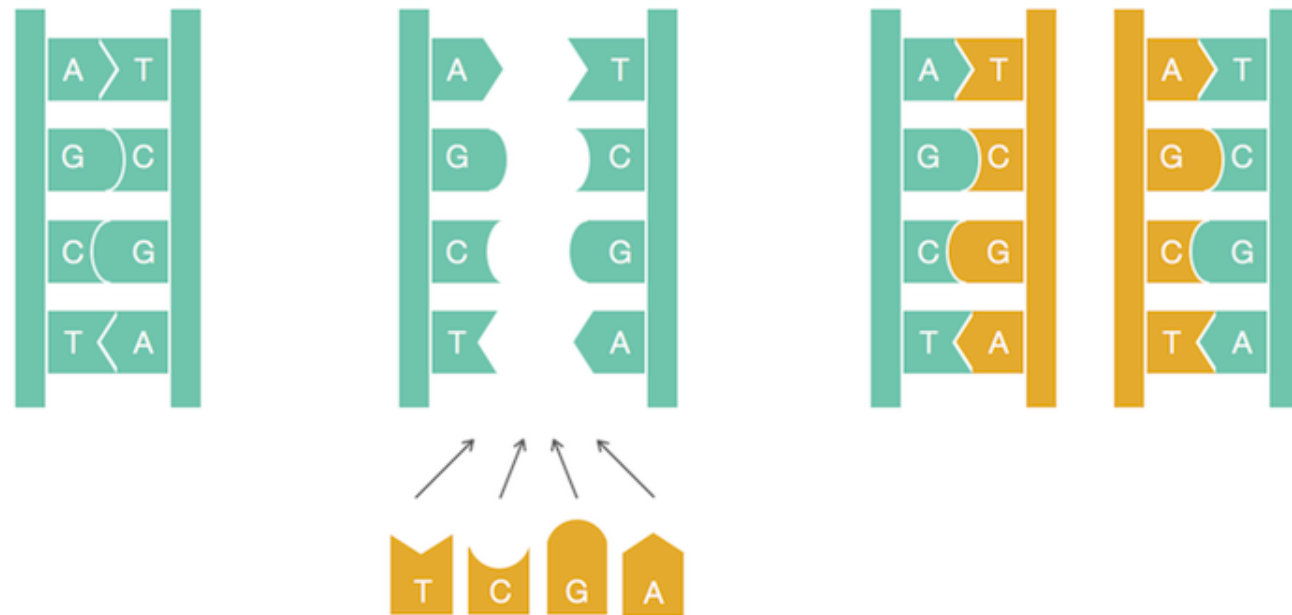
- Replication origin and DnaA boxes
- Frequent k-mers
- Finding regions of interest in the DNA
- Skew diagrams from biological knowledge
- Flexible patterns

Recap

- Biological data is sequential (DNA, RNA, proteins)
- Biological processes transform DNA->RNA->proteins
- Computers can process these sequences and simulate the biological processes

DNA replication

- `new_genome = genome.copy()` ??
- *“It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.”* Watson and Crick



- it's not that simple...

DNA replication

- Replication process can be analyzed with algorithms
- Identifying the replication origin (**oriC**) : useful for biomedical applications (gene therapy), agriculture, and other applications
 - We do not want to affect this region of the genome
- We will focus on bacterial genomes – much smaller (5Gbps)
 - Also we have more bacterial cells than our own cells
 - Circular – there is no beginning or end
- Problem: Given a circular bacterial genome, find the location of **oriC**
- Can we do this?

Finding oriC

- How does the cell know where to start?
- We need to find a “hidden message” in a known replication origin

```
atcaatgatcaacgtaagcttctaagcatgatcaaggtgctcacacagtttatccacaac
ctgagtggatgacatcaagataggtcgttgtatctccttcctctcgtactctcatgacca
cggaaagatgatcaagagaggatgatttcttgcccatatcgcaatgaatacttgtgactt
gtgcttccaattgacatcttcagcgccatattgcgctggccaaggtgacggagcgggatt
acgaaagcatgatcatggctggttctgtttatcttggttttgactgagacttgtagga
tagacgggtttttcatcactgactagccaaagccttactctgcctgacatcgaccgtaa
tgataatgaatttacatgcttccgcgacgatttacctcttgatcatcgatccgattgaag
atcttcaattgttaattctcttgccctcgactcatagccatgatgagctcttgatcatgtt
tccttaaccctctattttttacggaagaatgatcaagctgctgctcttgatcatcgtttc
```

- Some “words” might appear more often than others...

Hidden message

- Problem: find most frequent k-mers in a string
- **K-mer** – words of size k
- Given “ACA**ACTAT**GCATA**CTATC**GGGA**ACTAT**CCT” and k=5 ?
- COUNT(ACA**ACTAT**GCATA**CTATC**GGGA**ACTAT**CCT, ACTAT) = 3
- COUNT(CGATATATCCATAG, ATA) = 3 (overlapping occurrences of ATA)
- Replication is performed by DNA polymerase, but also need a protein called DnaA
- DnaA binds to short segments within replication origin (oriC) known as DnaA box
- DnaA boxes are our hidden message

Finding frequent words

- String dna has $|\text{dna}| - k + 1$ k -mers of size k
- Compute how many times a certain k -mer appears (Lab1 BA1A)
- Find the most frequent k -mers in a string (Lab1 BA1B)

```
FREQUENTWORDS(Text, k)  
  FrequentPatterns  $\leftarrow$  an empty set  
  for  $i \leftarrow 0$  to  $|\text{Text}| - k$   
    Pattern  $\leftarrow$  the  $k$ -mer  $\text{Text}(i, k)$   
     $\text{COUNT}(i) \leftarrow \text{PATTERNCOUNT}(\text{Text}, \text{Pattern})$   
   $\text{maxCount} \leftarrow$  maximum value in array  $\text{COUNT}$   
  for  $i \leftarrow 0$  to  $|\text{Text}| - k$   
    if  $\text{COUNT}(i) = \text{maxCount}$   
      add  $\text{Text}(i, k)$  to FrequentPatterns  
  remove duplicates from FrequentPatterns  
  return FrequentPatterns
```


Finding frequent words

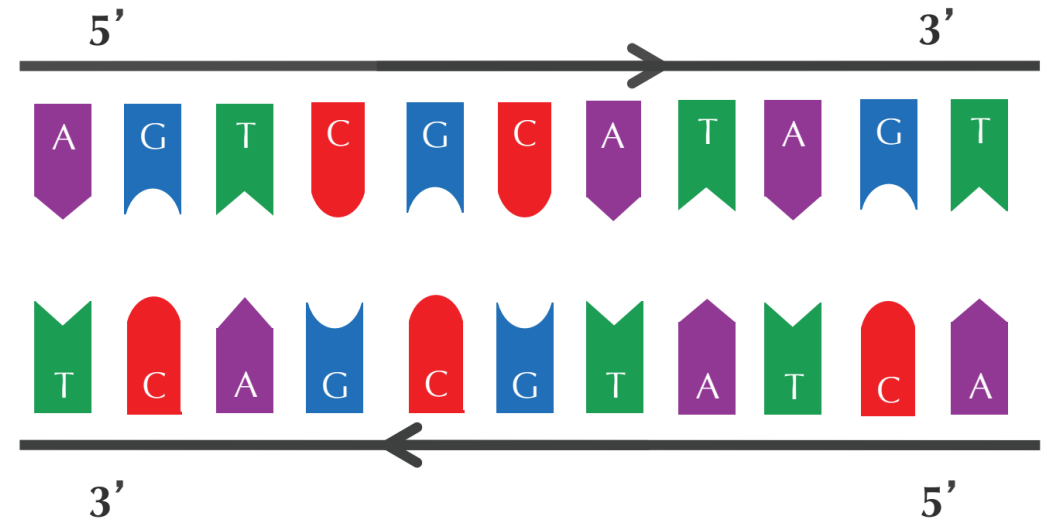
- For the previously shown *Vibrio cholerae*

<i>k</i>	3	4	5	6	7	8	9
count	25	12	8	8	5	4	3
<i>k</i> -mers	tga	atga	gatca tgatc	tgatca	atgatca	atgatcaa	atgatcaag cttgatcat tcttgatca ctcttgatc

Very unlikely
(1 in 1300):
Potential
DnaA
boxes?

Directionality in DNA strands

- A and T complement each other
- Same for C and G
- DNA string has two ends: 5' and 3'
- DNA is read 5'→3', so the complementary strand is also reversed (reverse complement – Lab BA1C)



Finding frequent words

- oriC region for the previously shown *Vibrio cholerae*

atcaatgatcaacgtaagcttctaagc**ATGATCAAG**gtgctcacacagtttatccacaac
ctgagtggatgacatcaagataggtcgttgtatctccttcctctcgtactctcatgacca
cggaaag**ATGATCAAG**agaggatgatttcttggccatatcgcaatgaatacttgtgactt
gtgcttccaattgacatcttcagcgccatattgcgctggccaaggtgacggagcgggatt
acgaaagcatgatcatggctgttgttctgtttatcttgttttgactgagacttgttagga
tagacgggtttttcatcactgactagccaaagccttactctgcctgacatcgaccgtaa
tgataatgaatttacatgcttccgcgacgatttacct**CTTGATCAT**cgatccgattgaag
atcttcaattgttaattctcttgccctcgactcatagccatgatgagct**CTTGATCAT**gtt
tccttaaccctctatTTTTTtacggaaga**ATGATCAAG**ctgctgct**CTTGATCAT**cgtttc

- DnaA protein does not care which strand it binds to
- But maybe these are just common k-mers along this genome and not just the oriC region?

Pattern Matching Problem (BA1D)

Pattern Matching Problem:

Find all occurrences of a pattern in a string.

Input: Strings *Pattern* and *Genome*.

Output: All starting positions in *Genome* where *Pattern* appears as a substring.

- For ATGATCAAG: 116556, 149355, **151913**, **152013**, **152394**, 186189, 194276, 200076, 224527, 307692, 479770, 610980, 653338, 679985, 768828, 878903, 985368
- Neither sequences appear elsewhere so close together (clump)

DnaA boxes

- Have we found the DnaA binding sequence for all bacteria?
- Not really, different bacteria have different DnaA boxes
- Frequent 9-mers (count>2) for *Thermotoga petrophila* oriC:

AACCTACCA

AAACCTACC

ACCTACCAC

CCTACCACC

GGTAGGTTT

TGGTAGGTT

- We know that the actual DnaA box is CCTACCACC/GGTGGTAGG, occurring 5 times in the oriC region
- What if we don't know oriC because it's a new organism?

Clump finding (BA1E)

- Look for windows of length L and check if a k -mer appears at least t times – an (L,t) -clump

Clump Finding Problem:

Find patterns forming clumps in a string.

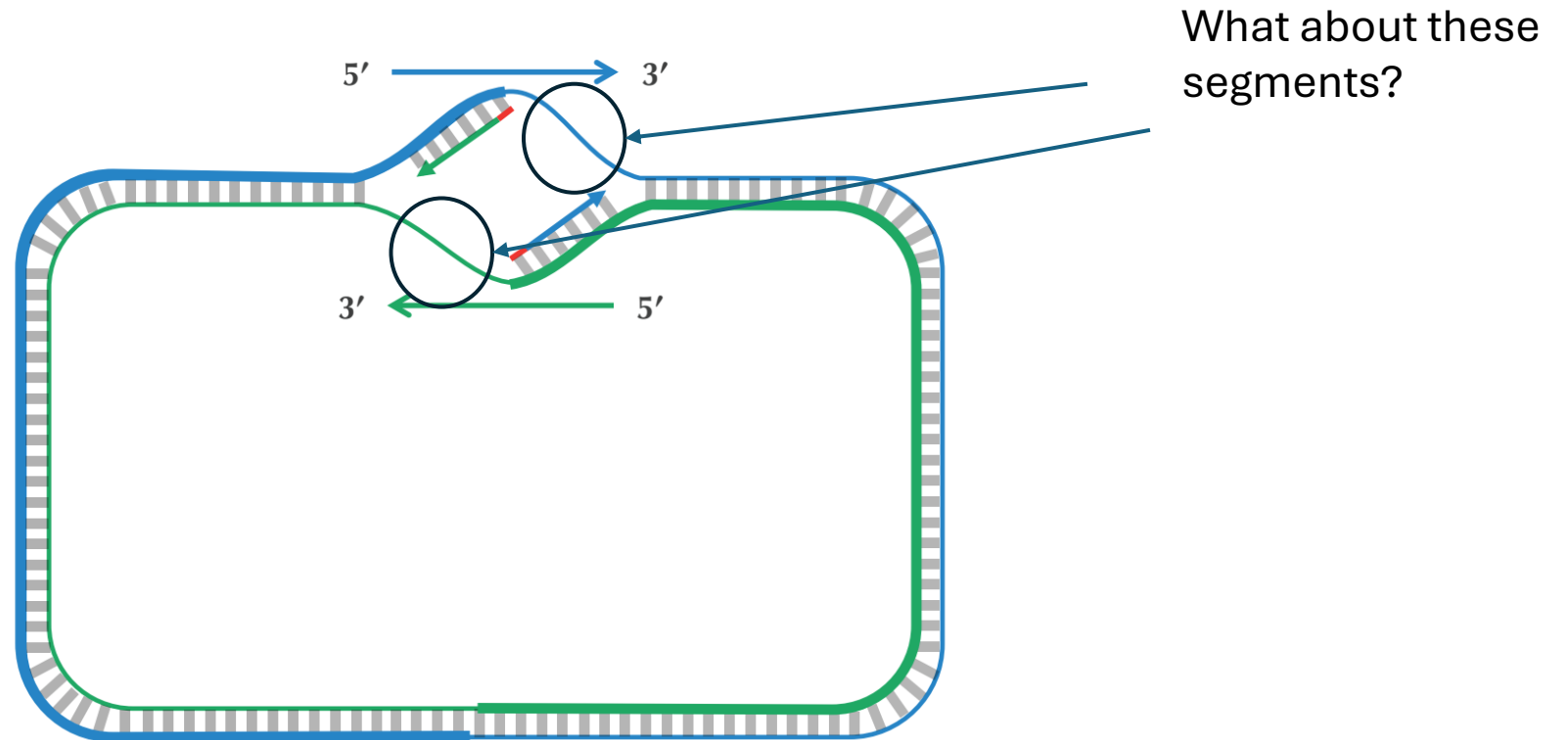
Input: A string *Genome*, and integers k , L , and t .

Output: All distinct k -mers forming (L,t) -clumps in *Genome*.

- For gatcagcataaggggtcc**CTGCAATGCATGACAAGCCTGCAGT**tggttttac
- TGCA is a $(25,3)$ -clump
- E. coli has hundreds of 9-mers forming $(500,3)$ -clumps
- What can we do?

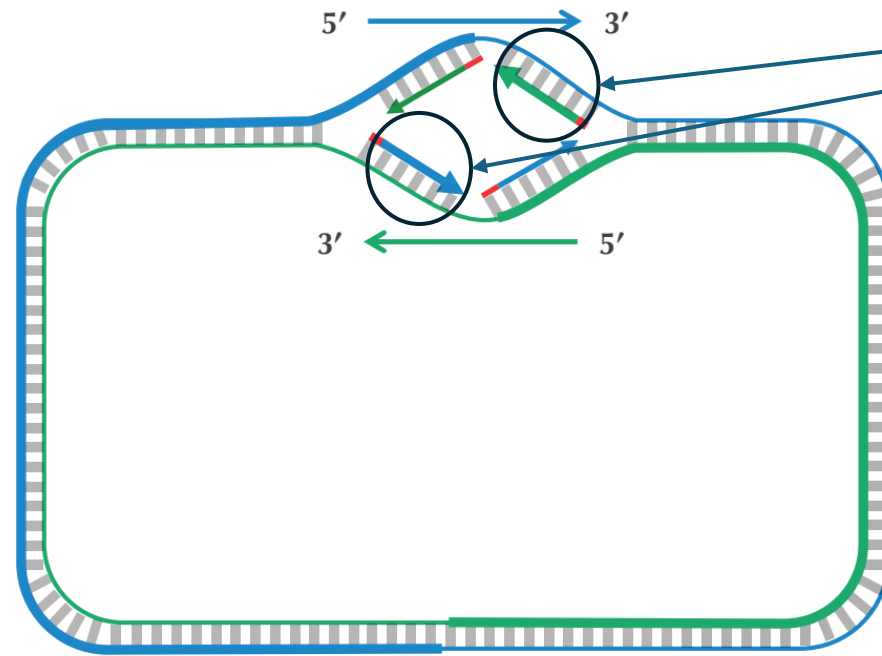
DNA replication is asymmetric

- DNA polymerase is unidirectional: it only runs 3'→5' (opposite direction of DNA)



DNA replication is asymmetric

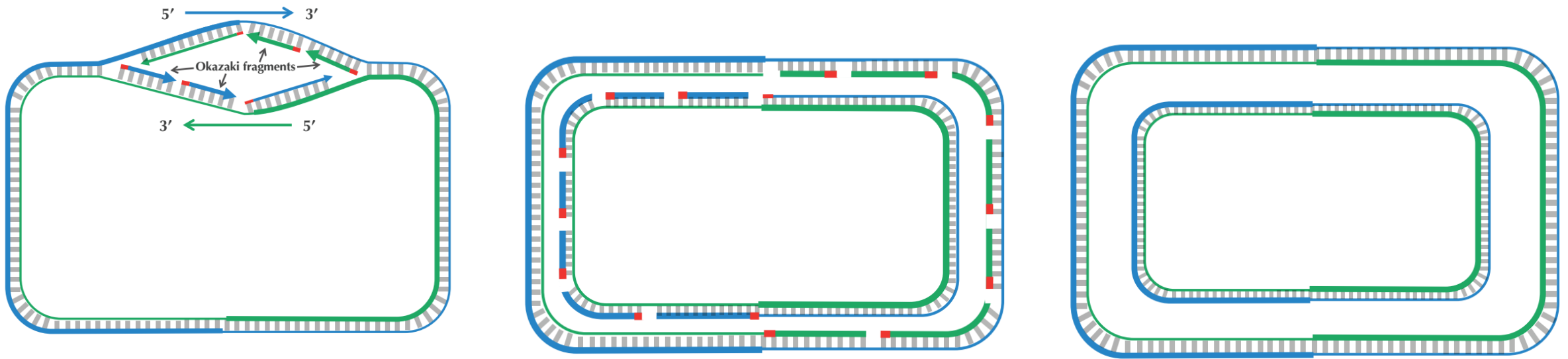
- DNA polymerase is unidirectional: it only runs 3'→5' (opposite direction of DNA)



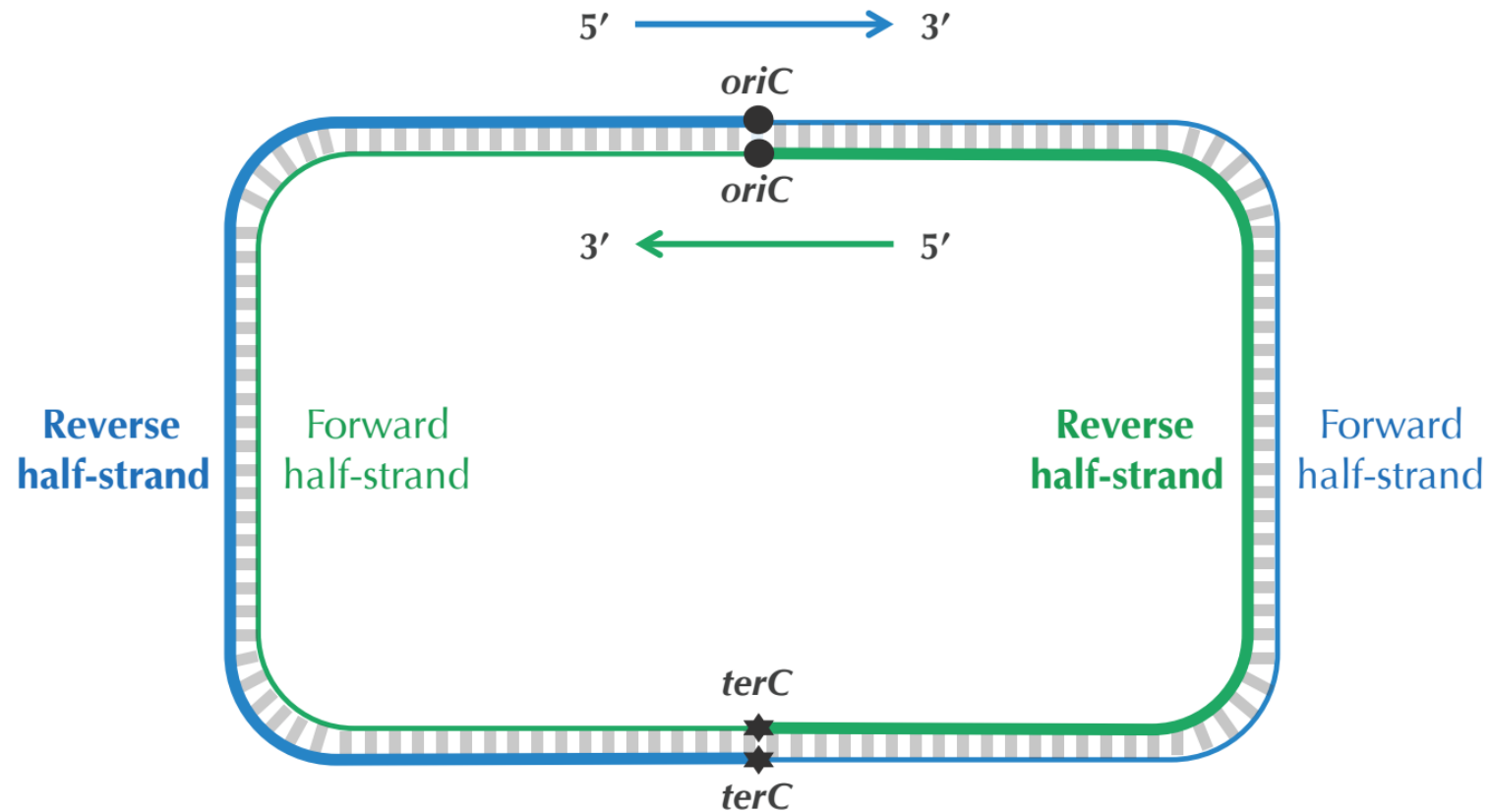
Must wait until the strand opens enough to replicate a small chunk of DNA

DNA replication is asymmetric

- Reverse (leading half-strand) is synthesized non-stop
- Forward (lagging half-strand) is synthesized in chunks and combined in the end



DNA replication is asymmetric



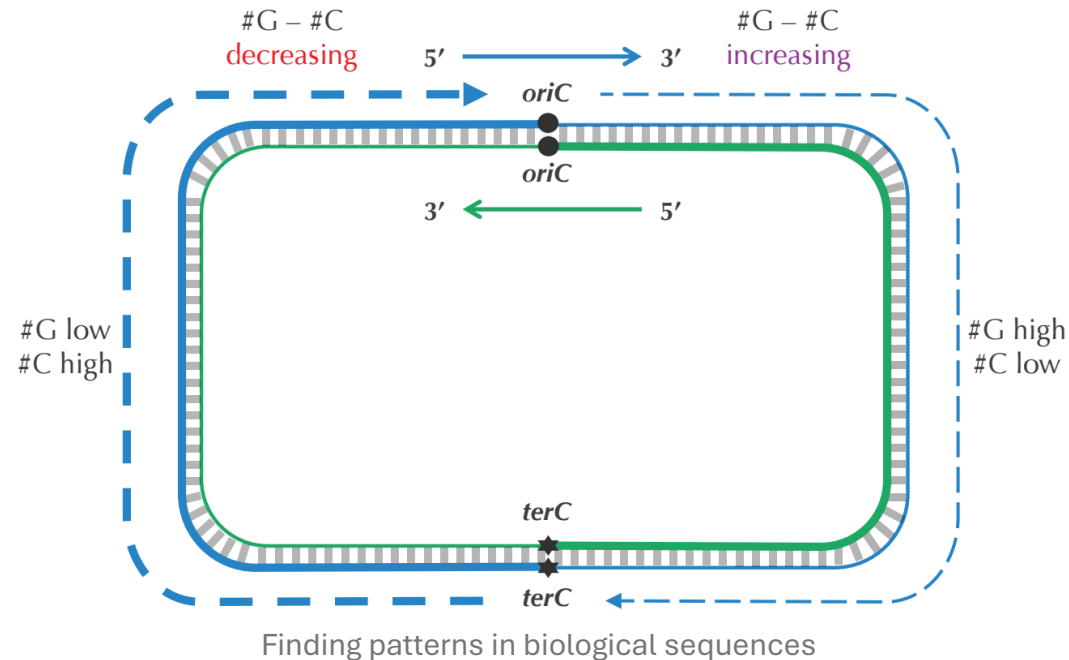
DNA replication is asymmetric

- Forward (lagging half-strand) is single-stranded for longer than the reverse (leading half-strand)
- Single-stranded DNA has a higher mutation rate: C has a tendency to mutate (leading to T-G mismatches, which are repaired to T-A)

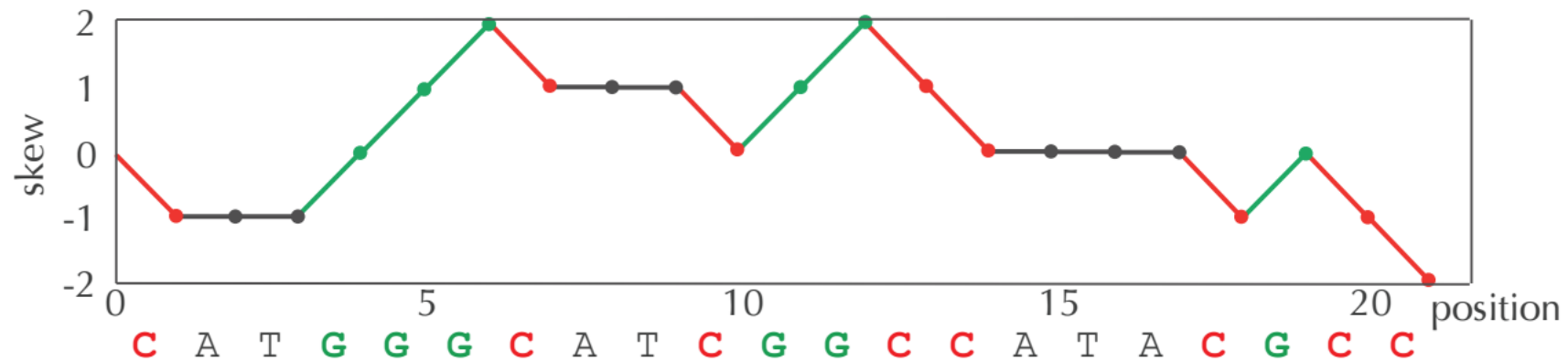
	#C	#G	#A	#T
Entire strand	427419	413241	491488	491363
Reverse half-strand	219518	201634	243963	246641
Forward half-strand	207901	211607	247525	244722
Difference	+11617	-9973	-3562	+1919

DNA replication is asymmetric

- If #G-#C is increasing – we are in the forward half strand
- If #G-#C is decreasing – we are in the reverse half strand
- What can we do with this information



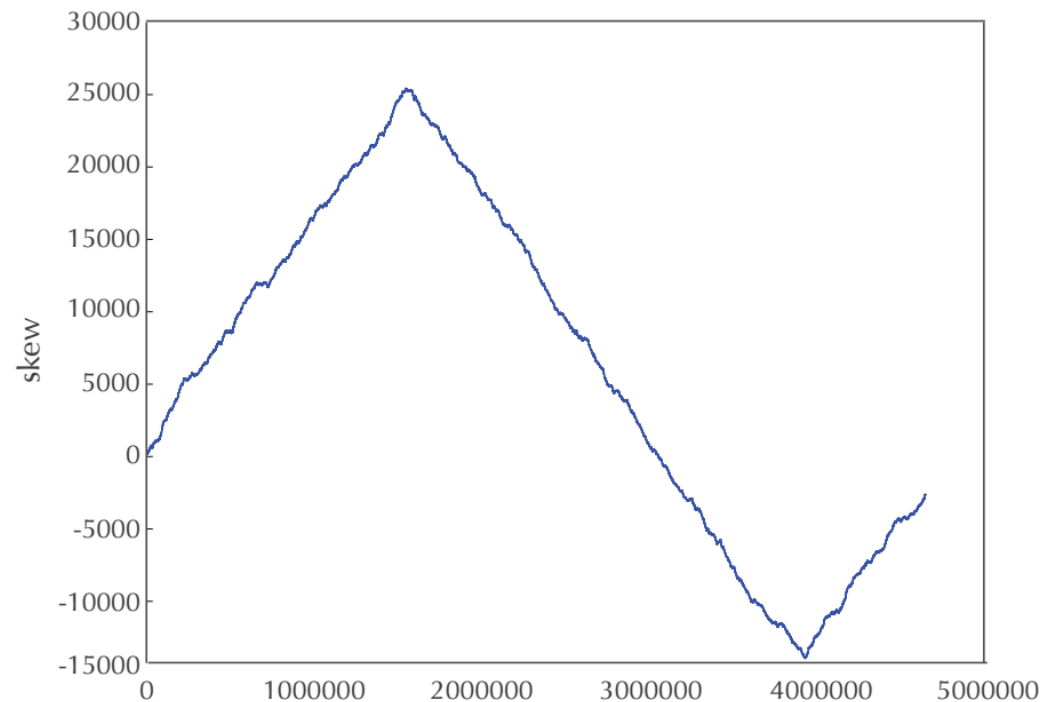
Skew diagram



- If we do this for the whole genome, and at some point $\#G - \#C$ goes from decreasing to increasing (i.e. $\#G$ starts increasing and $\#C$ starts decreasing)

oriC of *E. coli*

- What's the meaning of the position with the minimum skew?



Minimum Skew Problem (Lab1 BA1F)

Minimum Skew Problem:

Find a position in a genome where the skew diagram attains a minimum.

Input: A DNA string *Genome*.

Output: All integer(s) i minimizing $\text{SKEW}_i(\text{Genome})$ among all values of i (from 0 to $|\text{Genome}|$).

oriC of E. coli

- Now we where to look for DnaA boxes (around position 4M)
- We just have to find frequent 9-mers in this sequence...

```
aatgatgatgacgtcaaaaggatccggataaaacatggtgattgcctcgcataacgcggt
atgaaaatggattgaagccccggggccgtggattctactcaactttgtcggcttgagaaaga
cctgggatcctgggtattaaaaagaagatctat ttatttagagatctgttctattgtgat
ctcttattaggatcgcactgccctgtggataacaaggatccggcttttaagatcaacaac
ctggaaaggatcattaactgtgaatgatcggatgatcctggaccgtataagctgggatcag
aatgagggggttatacacaaactcaaaaactgaacaacagttgttctttggataactaccgg
ttgatccaagcttcctgacagagttatccacagtagatcgcacgatctgtatacttattt
gagtaaattaacccacgatcccagccattcttctgccggatcttccggaatgtcgtgatc
aagaatggtgatcttcagtg
```

- We couldn't find any!

oriC of E. coli

- What can we do now?
- Does it have to be a perfect match?
- *Vibrio cholerae* has some imperfect matches...

atca**ATGATCAAC**gtaagcttctaagc**ATGATCAAG**gtgctcacacagtttatccacaac
ctgagtggatgacatcaagataggtcgttgtatctccttcctctcgtactctcatgacca
cggaaag**ATGATCAAG**agaggatgatttcttggccatatcgcaatgaatacttgtgactt
gtgcttccaattgacatcttcagcgccatattgcgctggccaagggtgacggagcgggatt
acgaaag**CATGATCAT**ggctggtgttctgtttatcttgttttgactgagacttgttagga
tagacgggtttttcatcactgactagccaaagccttactctgcctgacatcgaccgtaa
tgataatgaatttacatgcttccgcgacgatttacct**CTTGATCAT**cgatccgattgaag
atcttcaattgttaattctcttgccctcgactcatagccatgatgagct**CTTGATCAT**gtt
tccttaaccctctattttttacggaaga**ATGATCAAG**ctgctgct**CTTGATCAT**cgtttc

Hamming distance (BA1G)

- We need to find these variations too: DnaA can bind to them too
- Calculate Hamming distance between two k-mers: number of position with different symbols (mismatches)

Hamming Distance Problem:

Compute the Hamming distance between two strings.

Input: Two strings of equal length.

Output: The Hamming distance between these strings.

Approximate Pattern Matching (BA1H)

- We can relax our pattern finding algorithm:
 - Approximate pattern matching: find matches of a k -mer in a sequence, with at most d mismatches

Approximate Pattern Matching Problem:

Find all approximate occurrences of a pattern in a string.

Input: Strings *Pattern* and *Text* along with an integer d .

Output: All starting positions where *Pattern* appears as a substring of *Text* with at most d mismatches.

Approximate Pattern Count (BA1I)

- We can also adapt our definition of most frequent k-mers to consider up to d mismatches:
 - Pattern that maximizes $\text{COUNT}_d(\text{Text}, \text{Pattern})$
 - The pattern itself does not have to occur if $d > 0$
 - E.g. **AAAAA** is the most frequent 5-mer with $d=1$ of **AACAAGCATAAACATTAAAGAG**

```
APPROXIMATEPATTERNCOUNT(Text, Pattern, d)  
  count  $\leftarrow$  0  
  for  $i \leftarrow 0$  to  $|Text| - |Pattern|$   
    Pattern'  $\leftarrow$  Text( $i, |Pattern|$ )  
    if  $\text{HAMMINGDISTANCE}(\text{Pattern}, \text{Pattern}') \leq d$   
      count  $\leftarrow$  count + 1  
  return count
```

oriC of E. coli

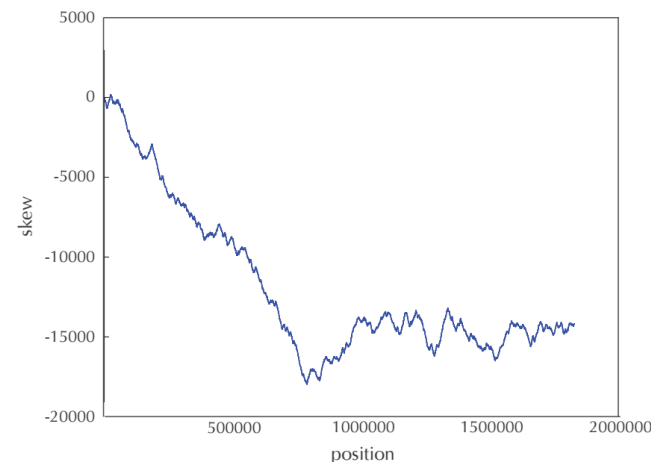
- Anything else to consider?
- Search of frequent words on the reverse complement too (Lab1 BA1J)
- Can we find the oriC of E. Coli now?
- We still need to try some windows of size 500 around the skew minimum but then...

```
aatgatgatgacgtcaaaaggatccggataaaacatggtgattgcctcgcataacgcggt
atgaaaatggattgaagcccgggcccgtggattctactcaactttgtcggcttgagaaaga
cctgggatcctgggtattaaaaagaagatctatatttagagatctgttctattgtgat
ctcttattaggatcgactgcccTGTGGATAAcaaggatccggcttttaagatcaacaac
ctggaaaggatcattaactgtgaatgatcggatgatcctggaccgtataagctgggatcag
aatgaggggTTATACACAactcaaaaactgaacaacagttgttcTTTGGATAAactaccgg
ttgatccaagcttcctgacagagTTATCCACAgtagatcgacgatctgtatacttattt
gagtaaattaacccacgatcccagccattcttctgccggatcttccggaatgtcgtgatc
aagaatgttgatcttcagtg
```

- Confirmed experimentally

oriC of E. coli

- However there are other 9-mer with the same frequency
 - GGATCCTGG, GATCCCAGC, GTTATCCAC, AGCTGGGAT, and CTGGGATCA
 - We don't know what is their purpose – could be regulatory DNA motifs
 - But getting with a short list of 6 candidates is already a great help
- Sometimes it is not that simple: skew plot of *Thermotoga petrophila*:



Conclusion

- Computational methods can help with uncovering biological mechanisms ...
- ... but need to be combined with biological knowledge to better inform the algorithms
 - E.g. we could only find candidates for the E. Coli DnaA box because we knew about the #G-#C skew
- Now you have the tools to find oriC and DnaA boxes on other bacteria
 - Can be a bit more complex in other types of organisms: archea, yeast, humans

Further reading

- Compeau Chapter 1